

PENGZHEN LI

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SUMMARY

Final-year Economics Ph.D. candidate (expected May 2027) specializing in applied econometrics, causal inference, and optimization. Four peer-reviewed publications (one with 1,000+ citations), six working papers including a job market paper on trade restrictiveness with James Lake, 2024 Holly Award for research and the 2026 Garrison Award for teaching. Build reproducible analytics in Stata/R/Python/SQL/GAMS and turn large administrative, trade, and production datasets into decisions. Authorized to work in the U.S. without sponsorship.

EDUCATION

University of Tennessee, Knoxville	Jan 2020 – May 2027 (expected)
Ph.D. in Economics — fields: international trade, applied econometrics, environmental and resource economics	
Shandong University	Sep 2016 – Jun 2018
M.A. in Finance — thesis published in <i>Technological Forecasting and Social Change</i> (1,000+ citations)	
Tongji University, Shanghai	Sep 2011 – Jun 2015
B.A. in Economics	

SKILLS

Programming / software: Python (pandas, numpy, statsmodels, scikit-learn), R, Stata, SQL, GAMS, MATLAB, ArcGIS, IMPLAN, Git, LaTeX.

Methods: causal inference (DiD, event study, IV, RDD, synthetic control, fixed effects), panel and time-series econometrics (GMM, panel threshold, TVAR, VECM, mGARCH), forecasting, mixed-integer and stochastic optimization, DEA, game theory (Shapley value, mean-field games), supervised and unsupervised machine learning, text-as-data.

Credentials / languages: CFA Level II; English (fluent), Mandarin (native).

RESEARCH EXPERIENCE

Doctoral Researcher , Department of Economics, University of Tennessee	2023 – present
<i>Trade restrictiveness under discriminatory tariffs</i> (with J. Lake; job market paper): derived welfare-based indices for partner-specific tariffs and a level/discrimination decomposition; built the WITS/Comtrade pipeline for 2017–2024 and quantified how MFN-only indices understate trade-war distortions. Presented at the Midwest Trade Meetings, 2025. <i>Multilateral tariff bargaining</i> (with G. Schaur): cooperative-game (Shapley value) model of coalition stability and gains from liberalization, calibrated to WTO bilateral flows for RCEP/CPTPP scenario analysis. <i>Mayoral endorsements and presidential elections</i> (with M. Van Essen, L. Lima): mean-field-game model of strategic endorsement timing tested on Brazilian and French administrative data; 2024 Holly Award.	
Graduate Research Assistant , University of Tennessee	Jan 2020 – Aug 2023
USDA NIFA and FAA ASCENT funded projects on sustainable-aviation-fuel supply chains.	
Built mixed-integer optimization models (GAMS) to design cost- and profit-optimal forest-residue	

sustainable-aviation-fuel supply chains across the Southeast U.S.; published in *Biomass and Bioenergy* (2026) and *Transportation Research Record* (2024).

- Ran facility-siting and regional impact analysis with IMPLAN and efficiency analysis with DEA; delivered scenario and sensitivity results (EMP, supervised ML, @RISK) to an interdisciplinary engineering-economics team and federal sponsors.
- Engineered reproducible ArcGIS/R/Python pipelines for high-resolution spatial data.

Earlier applied projects

2016 – 2020

- Panel-threshold evidence on green innovation and CO₂ (*TFSC* 2019, 1,000+ citations); synthetic-control estimate of the Wenchuan earthquake's long-run output effect; GMM analysis of MFP/CFAP payments and U.S. agricultural lending; TVAR/VECM/mGARCH commodity-price models; equity-premium prediction from *NYT/WSJ* text.

SELECTED PUBLICATIONS

“Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability.” **P. Li**, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

“Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA.” **P. Li**, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

“Beef price spread relationship with processing capacity utilization.” C. Martinez, **P. Li**, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

“Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data.” K. Du, **P. Li**, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303.

TEACHING

Instructor: ECON 351 Monetary Economics (Spring 2025, 2026); ECON 322 The Global Economy: Trade and Development (Summer 2025); ECON 313 Intermediate Macroeconomics (Summer 2026).

Teaching assistant: Ph.D. Microeconomic Theory and Mathematical Methods (ECON 511, 512, 581); Agribusiness Operations Research (AREC 525).

HONORS

Charles B. Garrison Award for Excellence in Teaching, University of Tennessee	2026
J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee	2024
J. Fred and Wilma Holly Fellowship (\$2,000 per year), University of Tennessee	2023 – present